

```

1.  %c0 = dataflow.addr[0] : {memShape = [32]} : [index] -> i32
2.  %0 = dataflow.load %arg0[%c0] : memref<32xi32> -> i32
3.  %1 = dataflow.load %arg1[%c0] : memref<32xi32> -> i32
4.  %2 = arith.add %0, %1 : i32
5.  dataflow.store %2, %arg2[%c0] : memref<32xi32> i32

```

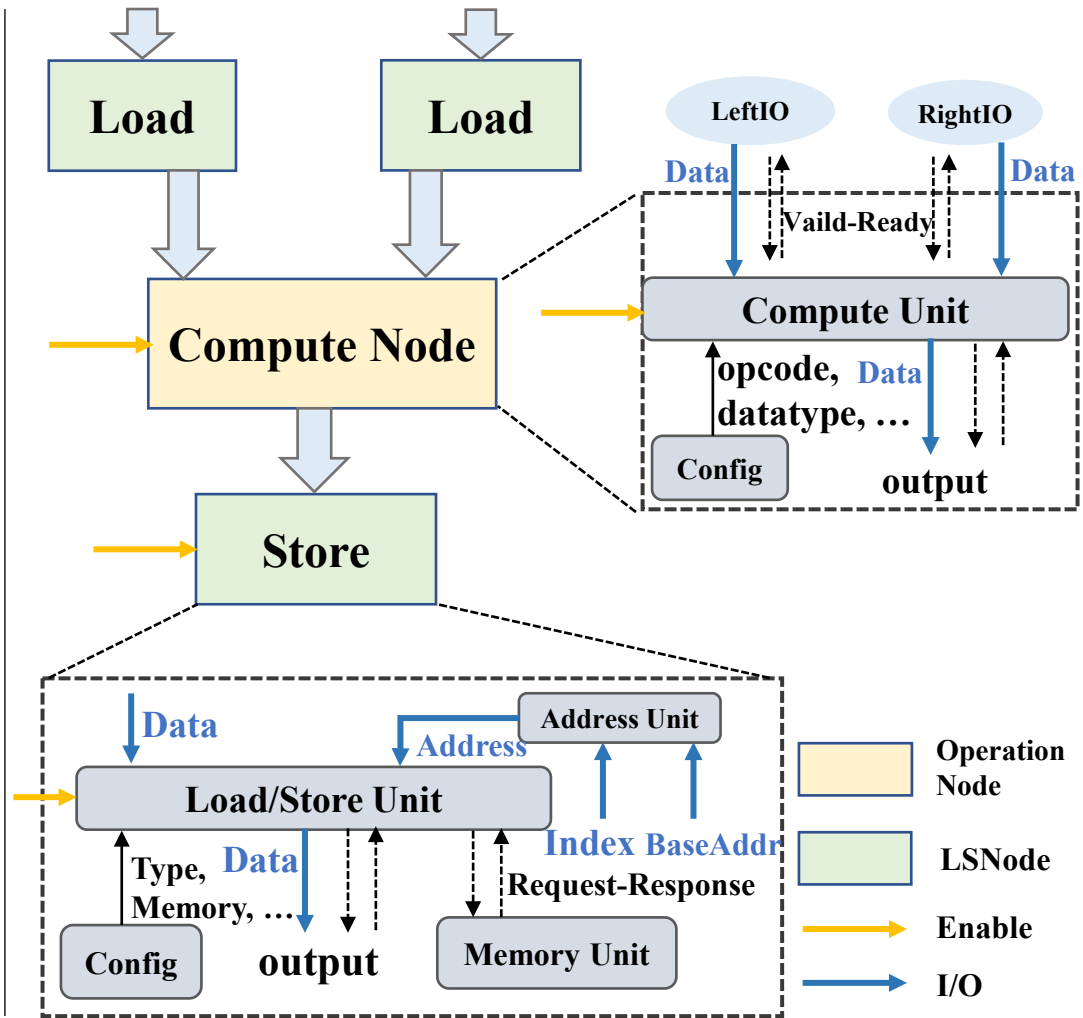
(a)

```

1.  // %c0 = arith.constant 1 : index
2.  val int_const_0 = Module(new ConstNode(value = 1, ID = 0))
3.  // %0 = dataflow.load %arg0[%0] : memref<32xi32>
4.  val load_2 = Module(new MemLoad(NumOuts = 1, ID = 2))
5.  // %1 = dataflow.load %arg1[%c0] : memref<32xi32>
6.  val load_3 = Module(new MemLoad(NumOuts = 1, ID = 3))
7.  // %3 = arith.addi %0, %1 : i32
8.  val int_add_4 = Module(new ComputeNode(NumOuts = 1, ID = 4, opCode =
  "Add"))(sign = false))
9.  // dataflow.store %3, %arg0[%c0] : memref<32xi32>
10. val store_5 = Module(new MemStore(ID = 5, RouteID = 0))
11. ...
12. int_add_4.io.LeftIO <> load_2.io.Out(0)
13. int_add_4.io.RightIO <> load_3.io.Out(0)
14. store_5.io.inData <> int_add_4.io.Out(0)

```

(b)



(c)